



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II VMS Table Publication – Variable Message Signs

Release 1.1rev0

National Traffic Information Centre

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1	Introduction	3
1.1	General terms.....	3
2	Concepts.....	4
2.1	Structure of a static publication	4
2.2	Concept of publication structure	4
2.3	Types of described signs.....	4
3	Other important aspects.....	6
3.1	VMS configuration.....	6
3.2	VMS identification and versioning	6
3.3	Information lifecycle in a static publication	7
3.3.1	VmsController structure life cycle.....	7
3.3.2	Life cycle of the VmsControllerTable structure.....	7
3.3.3	Other	7
3.4	References to other tables	7
3.5	References to graphical representations (images).....	8
3.5.1	Graphical representation of the controller	8
3.5.2	Graphical representation of VMS.....	9
4	VMS location and position description concepts.....	10
4.1	Change of location (position)	10
4.2	Location, position on the portal and affected lanes	10
4.3	Geographical description of the VMS location.....	11
4.4	Position of the display area within the VMS	11
5	Elements of a static publication	13
6	Changelog.....	18

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- **Publisher** NDIC
- **Author** Petr Bureš

This Specification defines data format for publishing information about variable message signs along roads for information and traffic management purposes. The profile has a static and a dynamic part.

The static part (this document) describes the basic characteristics of the traffic sign/portal including the configuration of the display area (portal), position and validity relative to the lane.

The dynamic part describes the information currently displayed on the individual signs (on the portal), including in the case of mobile trailers the configuration change of the displayed area and dynamic information about its position.

Version history

See Changelog chapter.

1 Introduction

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NTIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NTIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Format description of data structure for transferring some kind of information

Protocol description of data transfer process and rules

NTIC national traffic information centre

DATEX II model class model for describing information related to the road traffic. The model in form of UML, is platform independent. One of specific platforms for expressing the model is XML. The model is extensive (it consists of few thousands of structural items representing term incl. Definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [d2ref]. For additional information see [d2supp].

(DATEX II) profile simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

Publication an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example `SituationPublication` [16157-3], `ElaboratedDataPublication` [16157-5], `PredefinedLocationsPublication` [16157-3] etc.

(W3C) XML schema language, which allows definition of rules for a XML document structure.

XSD XML document containing XML schema. Sometimes it is synonym for XML schema itself.

Uri uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

Portal a structure above or along the road on which signs are mounted (gantry, cantilever)

2 Concepts

2.1 Structure of a static publication

The static publication has the following structure:

- **VmsControllerTable**: multiple signs (controllers) grouped by administrator / by type
- **VmsController**: one logical unit (portal, cantilever, sign along the road, trailer)
- **Vms**: specific indexed part of the portal (controller), including location
- **DisplayArea**: indexed part of Vms of pictogram or text type

2.2 Concept of publication structure

- Each portal/cantilever/separate sign/information sign/trailer is described separately by the "**VmsController**" structure.
- The "**VmsController**" structures are categorized in the static publication into 3 **vmsControllerTable** tables containing:
 - Dynamic permanent (Along the road, On the portals, cantilevers) signs,
 - Dynamic portable (mobile trailers, movable) signs
 - Fixed static (unchanging "metal" signs) signs
- The "**VmsController**" structure has as many separate "**Vms**" elements as there are separate physical elements on the portal.
 - Information portals, trailers containing one text panel and/or one or more pictogram panel are modelled as one "**Vms**"
 - Line control portals are modelled as multiple "**Vms**".
- Each "**Vms**" has as many "**DisplayArea**" elements as there are logically separated display areas present on one physical structure.
- Each "**DisplayArea**" describes either:
 - Pictogram (**PictogramDisplayArea**) or
 - Text information (**TextDisplayArea**).
- Supplementary tables are not modelled in the static publication. They are addressed directly in the dynamic publication as part of a Pictogram structure.

2.3 Types of described signs

In the publication, for the sake of sorting at the level of individual VMS, we distinguish between different types of traffic signs (using the optional text **vmsTypeCode**) into:

- **mobileVariableMesageSign**: different combinations of pictograms and textual displays on the trailer according to the trailer configuration
- **variableInformationPanel**: combination warning pictograms (PDZ) [0-2] and textual displays (PIT) [1]
- **variableMesageSign**: one warning, command, prohibition pictogram (PDZ) with an optional supplementary panel.
- **variableDirectionSign**: variable continuous (not digital) directional traffic sign, a panel along the road with information about the further direction of the road

- temperature: special type of “variableInformationPanel” (text) with two lines and fixed meaning for displaying temperature information (1st line ambient temperature and 2nd line road temperature in degrees C)
- parking: special type of “variableInformationPanel” (text) with usually one or two lines with information about parking area occupancy. 1st line displays information about closest parking area, 2nd line about farther parking area.
- metalSign: any non-variable sign, even directional information, custom, ...can be part of variable portals.

3 Other important aspects

3.1 VMS configuration

- Each non-moving "**Vms**" contains directly in the "**vmsLocation**" element a description of the location to the level of the affected lanes and a description of the position along the road.
- The individual **DisplayAreas** within a "**Vms**" have an exact position on the Vms using relative and absolute positional directions.
- When the sign configuration is permanently changed, a new version of the description is created at the static publication level, the old version is preserved.
- Temporary changes to the configuration of displays within one Vms (e.g. one description in the morning and another in the evening) are handled in the dynamic publication using "**vmsConfigurationOverride**" element.
- Logical elements (**true/false**) are used for both values. The absence of an element does not imply the value "false" only that the state is unknown.
- Elements with enumerations are included only if they provide information value.

3.2 VMS identification and versioning

Identification is handled at several levels within the publication. A portal (controller), a traffic sign, is identified by the VMS table (**VmsControllerTable**, id and version) it is in and by its own identifier (**VmsController**, id and version). The individual parts of a portal (controller), traffic sign (**Vms**, index) are identified by the index related to that sign, portal. The individual display areas on one part of a sign (**DisplayArea**, index) are identified by the index related to that part of the sign.

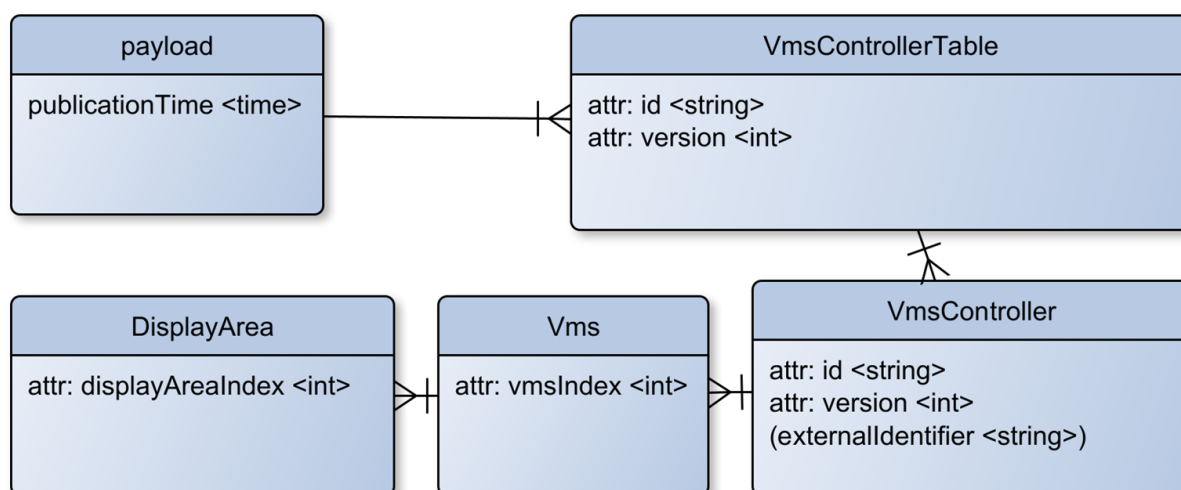


Fig 1. Identifier structure (own source)

structure	Description of identification	Example
VmsControllerTable	Unambiguous identification of the table using unique id and version. Recommendation: use the type of controllers in the table	id="CzechVmsTable-mobileVms" version="1"

VmsController	Unambiguous identification of the structure using a unique id and version. Recommendation: use id (shortId) Alternative: use portal label according to "R120 Portal Registration Numbers" for mobile trailers use e.g.: MobileVMS_number_type, where type can be e.g. full matrix = FM, simple matrix = SM	id="3af3c14996e54" version="1"
Vms	Identification within a single parent structure using an index, always starting from 1	vmsIndex="1"
DisplayArea	Identification within a single parent structure using an index, always starting from 1	displayAreaIndex="1"

3.3 Information lifecycle in a static publication

3.3.1 VmsController structure life cycle

- New versions of **VmsController** (identified by the same id and one higher version number) are added only in case of permanent changes (> 6 months) of controller configuration, display configuration or location / position.
- The original versions of **VmsController** remain in the table to maintain backwards compatibility and the ability to reference old events.
- The **VmsControllerTable** table number is not changed by the above changes.

3.3.2 Life cycle of the VmsControllerTable structure

- The new version of **VmsControllerTable** is created only in important cases when the tables are purged of previous, no longer needed "ballast".
- The original version of the **VmsControllerTable** is no longer changed and remains available through a separate publication.

3.3.3 Other

- Any short-term changes to the display configuration, lane changes or changes of the location of the trailer are part of the dynamic publication (without the new version).
- Only one version of **VmsController** can be valid at a time, the process of activation and deactivation of versions is described in the dynamic publication.

3.4 References to other tables

- Pictograms or texts displayed on Vms can be specified beyond the DATEX II publication using predefined tables. In a static publication, these tables are referenced using the structures: pictogramCodeListIdentifier and textCodeListIdentifier.
- These tables are for explanatory purposes only (i.e. how a certain symbol was encoded) or as a phrase translation table in case of repetitive textual description on the textual panel. It is up to the subscriber to evaluate their suitability.
- Tables related to this profile

- Pictogram table: RSD_Mapping.xlsx
- Table of text phrases: none

3.5 References to graphical representations (images)

The static publication refers to the images of each VMS using the "**Vms > imageLink**" structure. Furthermore, the concept of using the "**VmsController > externalIdentifier**" element (from the ASFINAG profile) is adopted for the display of the entire controller (portal).

The following structure is used to reference the images:

- baseUrl/customUrl/vms/VmsControllerId/VmsId/DisplayAreaId

3.5.1 Graphical representation of the controller

The graphical representation of the controller (in the "**VmsController > externalIdentifier**" element) contains:

- Basic type configuration of the controller, a **static image** showing the position of the VMSeS within the controller and the display areas on the VMS.
- Includes any other non-variable parts of the sign that are important to understand the sign meaning (for example, for a temperature or parking VMSeS, or surrounding sign symbols on a trailer).
- Can be an annotated real picture or a schematic representation.

Example of the reference "**VmsController > externalIdentifier**"

- <https://rsd.cz/cache/vms/3af3c14996e54> = controller image

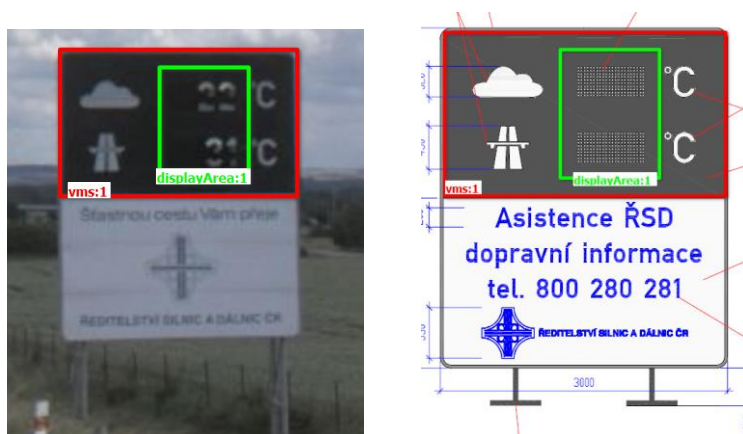


Fig 2. Controller display with temperature information, source: (own, control system R19)

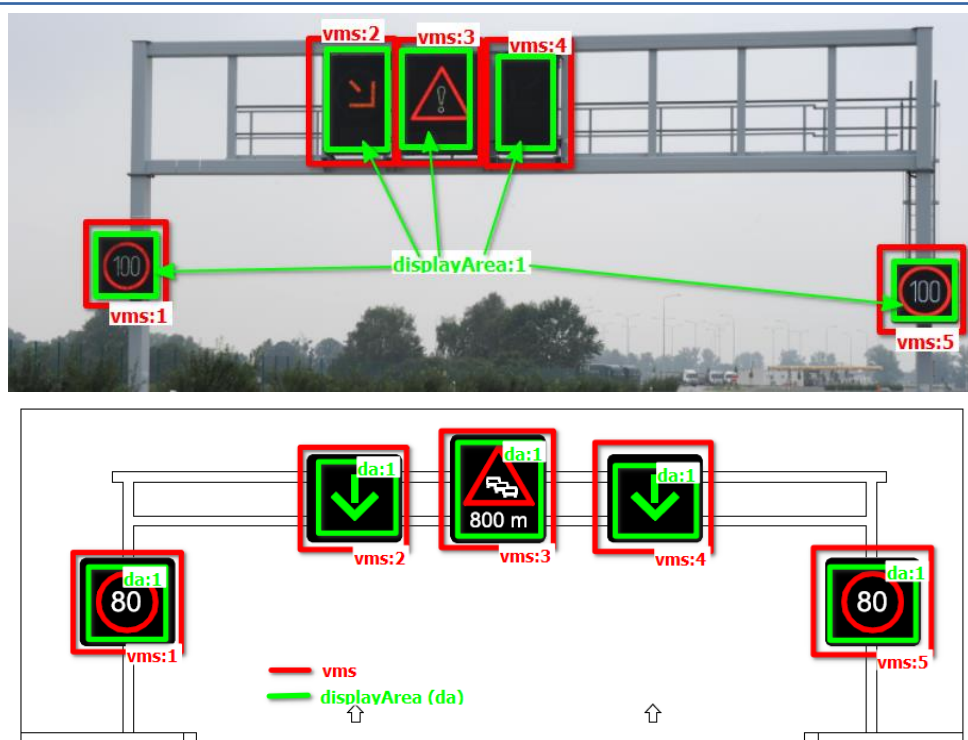


Fig 3. Display of controller with linear control, source: (own, ŘSD TP 141)

3.5.2 Graphical representation of VMS

The graphical representation of VMS (in the "**Vms > imageLink**" element) contains:

- **Actual display of** the VMS data including all display areas (DisplayArea) and supplementary information panels

Example of the link "**Vms > imageLink**"

- <https://rsd.cz/cache/vms/3af3c14996e54/01> = picture of 1st VMS on the controller

4 VMS location and position description concepts

- Location and position description of fixed signs is always in **static publication**.
- The location and description of position of trailers and portable signs (i.e. for temporary lane restrictions) is in **dynamic publication**.
- Description of the sign location and position on the portal and along the road and affected lanes is in the element "**Vms > vmsLocation**".
- **This profile does not use "Vms>managedArea"** to describe the affected area (road).

4.1 Change of location (position)

- Location changes related to lane realignment, trailer moved to a new location, or a moving trailer are completely handled in the dynamic publication by **vmsLocationOverride** class.
- In the case of a more permanent placement of the trailer (> 6 months), a new version with an updated location description can be created to optimize the transmitted data.
- Handling of the original versions is described above in the "[life cycle](#)" section.

4.2 Location, position on the portal and affected lanes

- The location of the sign (at the VMS level) can be:
 - roadsideMounted: along a road on a post
 - roadsideCantileverMounted: on the semi-portal
 - gantryMounted: on the portal
 - trailerMounted: on a trailer
 - centralReservationMounted: on the central reservation
 - overheadBridgeMounted: on the overhead bridge
 - tunnelEntranceMounted: on the tunnel entrance
 - vehicleMounted: on the vehicle
- The VMS location including name / number and direction of the road is in the **SupplementaryPositionalDescription**, (in other descriptions if not necessary is this information is omitted)
- The **position** of the VMS in relation to the lane / carriageway is specified in the element "**SupplementaryPositionalDescription > positionOnCarriageway**":
 - onTheLeft: on the post (of a portal) on the left in the direction of travel
 - inTheCentre: on the gantry above the road (also applies to cantilevers)
 - onTheRight: on the post (of a portal) on the right in the direction of travel
- **The validity** of the VMS with respect to the lane affected by the sign is specified in the element "**SupplementaryPositionalDescription > Carriageway**":
 - Carriageway: mainCarriageway = on the main carriageway
 - Then one of the following:
 - laneUsage: allLanesCompleteCarriageway = applies to all lanes, or
 - laneNumber: 0=hard shoulder ... n, for specific lane affected
- The relative position of the VMS placed on the portal (and its posts) results from the order of VMS numbering from left -> right, top -> bottom. (see Figure 4).

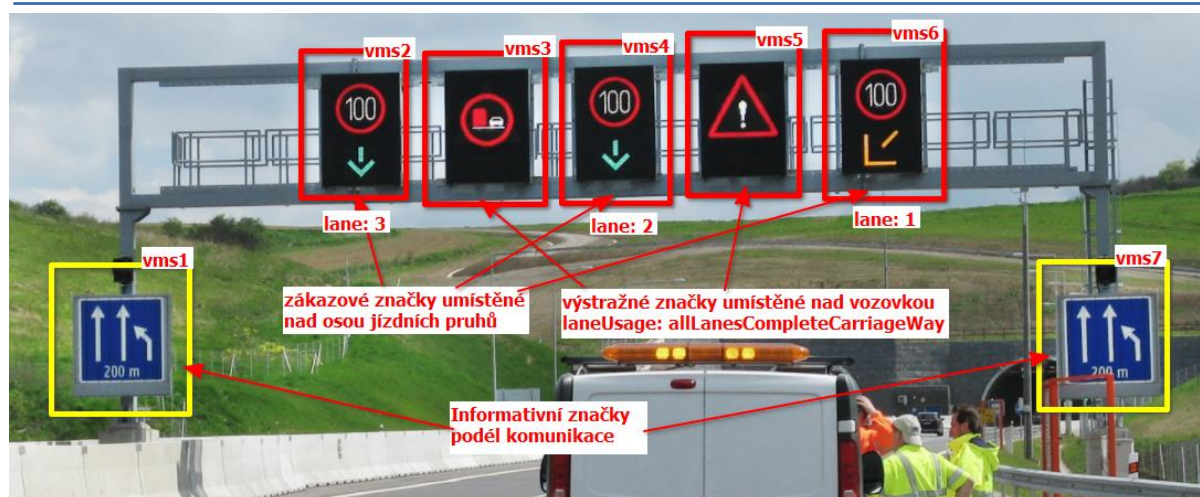


Fig. 4: Position and validity of the VMS in relation to lanes (source: ŘSD, modified)

4.3 Geographical description of the VMS location

- The location is always described by the **pointLocation** class.
- The location, described by the coordinates (WGS84), is given by the centre line of the carriageway and perpendicular line to VMS structure position along the road.
- Description of direction, name and road number always in the supplementary description
- Classes used to describe the position of the Vms are:
 - PointByCoordinates
 - PointAlongLinearElement (2x)
 - pointAlongLinearElement: GlobalNetwork as referent
 - pointAlongLinearElement: by mileage as referent
 - OpenlrPointLocationReference (optional)

4.4 Position of the display area within the VMS

- DisplayAreas are numbered from top to bottom and left to right within the VMS.
- The individual **DisplayAreas** within a "Vms" have their position on the Vms determined by relative and absolute position descriptors.
- The position of the first DisplayArea in the VMS is fixed by absolute position descriptors, other areas use relative descriptors (use area with lower index as a reference).

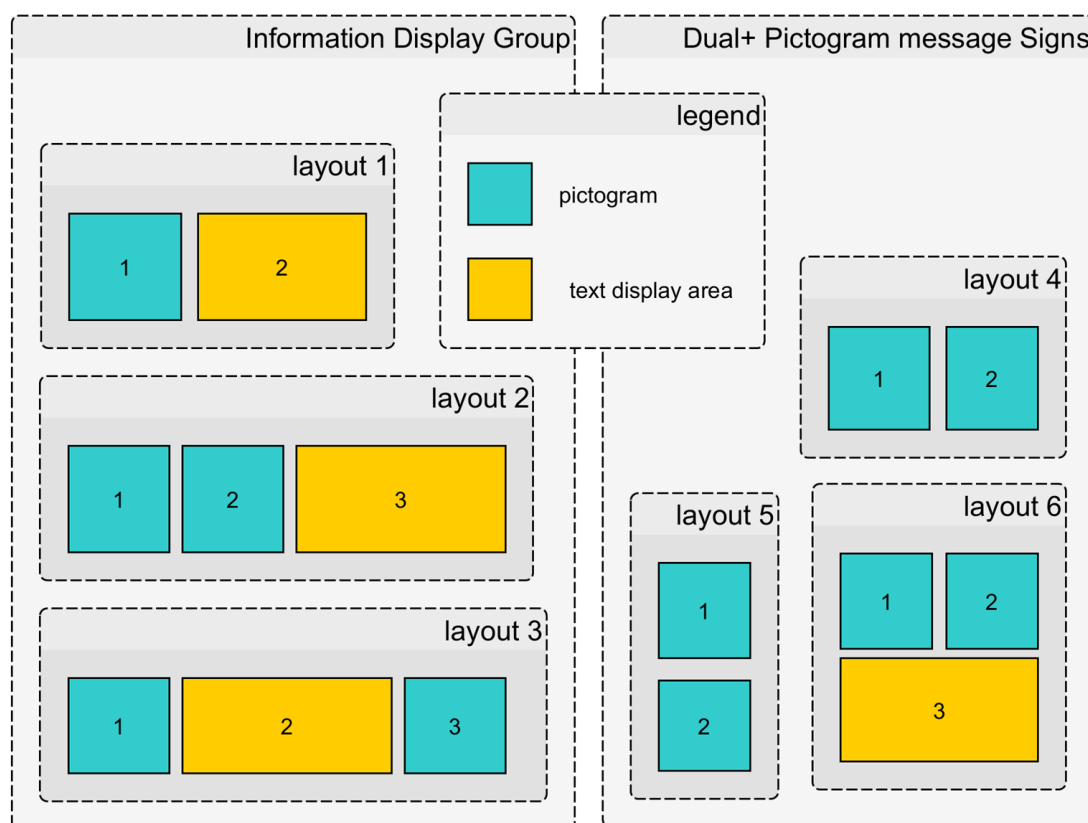


Fig. 5: DisplayArea layout within VMS

nayout	DisplayArea 1	DisplayArea 2	DisplayArea 3
1+4	positionXAbsolute:onLeft, positionYAbsolute:inTheMiddle	positionXRelativeToPrevious:toTheRight, positionYAbsolute:inTheMiddle	
2+3	positionXAbsolute:onLeft, positionYAbsolute:inTheMiddle	positionXRelativeToPrevious:toTheRight, positionYAbsolute:inTheMiddle	positionXRelativeToPrevious:toTheRight, positionYAbsolute:inTheMiddle
5	positionXAbsolute:inTheMiddle, positionYAbsolute:atTop	positionYRelativeToPrevious:alignedOn TheBottomSide	
6	positionXAbsolute:onLeft positionYAbsolute:atTop	positionXRelativeToPrevious:toTheRight, positionYAbsolute:atTop	positionXAbsolute:inTheMiddle positionYAbsolute:atBottom

5 Elements of a static publication

For some elements the cardinality (card.) has been changed so that they are always present (change marked **in green**) if the change is not possible due to higher variability, at least the desired state is marked **in purple**, most of enumerations are reduced to contain just values really used in publication (used enumerations values are marked **in orange** to distinguish them from examples).

Element payload [1]

element	card.	content/example
@ attributes of the payload element	1..n	Model Version, Publication Type, Namespace Definitions, Publication Language and Schema Location
feedDescription\values\value[@lang=cs]	0..1	x-source:cz-ndic_d2-vms-table-fixed-v1.1 ; the source URI in the traffic information register (https://registr.dopravniinfo.cz/)
feedType	1 ..1	x-format:cz-ndic_d2-vms-table-v1.1 the format URI in the traffic information register (https://registr.dopravniinfo.cz/)
publicationTime	1	2021-12-22T10:45:00+02:00; file publication time
publicationCreator\country	1	cz ; country of the publication provider
publicationCreator\nationalIdentifier	1	NDIC ; abbreviation of the name of the publication provider
headerInformation\confidentiality	1 ..1	noRestriction ; always this, providing without restrictions
headerInformation\informationStatus	1	real ; always this, providing only real information
vmsControllerTable	1..n	structure ; allowing multiple VMS controller tables to be described, separated by controller type (see @id below).

Element vmsControllerTable [1..n]

element	card.	content/example
@id	1	Allowed variants: "CzechVmsTable-fixedVms"=Dynamic fixed (Along the road, On the (semi)portal), "CzechVmsTable-mobileVms"=Dynamic portable. (trailers) "CzechVmsTable-metalSigns"=Fixed static ("metal" signs)
@version	1	1; table version, the version is incremented only in the case of significant interventions and table clean-up, simply adding or changing a controller does not trigger a new table version.
vmsControllerTableIdentification	0..1	2022-02-05; information about update of the table, can be date or any other descriptive information including "minor versions".
informationManager\country	0..1	cz ; country of residence of the creator of the publication content
informationManager\nationalIdentifier	0..1	NDIC ; abbreviation of the name of the creator of the content of the publication
vmsController	1..n	structure ; description of a single controller, e.g., a line-controlled portal or variable text panel, a mobile trailer, or a variable message sign along a roadway.

Element vmsController [1..n]

element	card.	content/example
@id	1	"3af3c14996e54" unique identifier

@version	1	1; version of the controller, with each change inside it is incremented
numberOfVms	1..1	2; number of physical VMS on portal, post
externalIdentifier	1..1	https://rsd.cz/cache/vms/3af3c14996e54 ; Basic configuration of the controller, static image with depicted areas for additional graphics (Vms and DisplayArea) Contains all other unchangeable parts of the sign important for its understanding (for example, for temperature VMS or parking navigation VMS, or surrounding symbols of the sign on the trailer)
vms	1..n	structure ; allowing to describe a single VMS consisting of one or more display areas.

Element vms [1..n]

element	card.	content/example
@vmsIndex	1	1; sequential index of VMS on the controller (always from 1 for each VMS), numbered from left to right and top to bottom
lanternsPresent	1..1	true; if separate flashers are present or a piece of the VMS matrix is dedicated to them
owner	0..1	multilingualString=ŘSD ČR; owner of the sign
physicalSupport	1..1	Allowed variants: roadsideMounted : along the road on a post roadsideCantileverMounted : on the semi-portal gantryMounted : on the portal trailerMounted : on the trailer centralReservationMounted : on the central reservation overheadBridgeMounted : on the overhead bridge tunnelEntranceMounted : on the tunnel entrance vehicleMounted : on the vehicle
vmsType	1..1	Allowed variants: fullMatrixSign : fully programmable matrix monochromeGraphic : monochrome display (temperature, parking) simpleMatrixSign : standard VMS with predefined options other : for the metal signs
vmsTypeCode	1..1	Selected sign resolution at VMS level mobileVariableMessageSign : combination of pictogram and text area on the trailer variableInformationPanel : combination of warning pictogram (0-2) and text area (1) variableMessageSign : one pictogram with supplementary panel variableDirectionSign : variable continuous (non digital) directional traffic sign, temperature : special text panel with two lines and fixed meaning (about temperature) parking : special text panel with two lines and fixed meaning (about parking) metalSign : any non-variable sign
dynamicallyConfigurable	1..1	true ; if the VMS area is dynamically configurable, true only for the fullMatrixSign type otherwise false
heightAboveCarriageway	0..1	5.6 ; height in meters above the road, for signs on the portal above the road, omitted for signs on posts along the road
dedicatedUsage	0..1	Allowed variants (otherwise omitted): laneControlSystem : for obligation and prohibition signs in the portal parkingInformation : for navigation to the parking lot tunnelManagement : for tunnel management (inside and outside)
vmsConfiguration	1	structure ; to describe the configuration of individual display areas.
vmsLocation	0..1	structure ; to describe the location of a VMS, for each VMS separately including supplementary description, the location uses the pointLocation type and is further described by a point (coordinates) and a position along the communication (mileage

		and GlobalNetwork as referent) and optionally by the OpenLR structure. It is not used for trailers in the static publication (depot position can be used).
imageLink\urlLinkAddress	1..1	https://rsd.cz/cache/vms/3af3c14996e54/01 ; Actual display on the VMS including all display areas (DisplayArea) and supplementary information panels

Element vmsConfiguration [1..n]

element	card.	content/example
numberOfDisplayAreas	1..1	1; number of active display areas per VMS
displayArea	1..n	structure; to describe the configuration of individual display areas.

Element displayArea [1..n]

element	card.	content/example
displayArea	1	structure; abstract element encapsulating general information about the displayed area. Allowed PictogramDisplayArea and TextDisplayArea types

Element TextDisplayArea [1..n]

element	card.	content/example
positionXAbsolute	0..1	onLeft , inTheMiddle , onRight = horizontal position relative to the VMS edge
positionYAbsolute	0..1	atTop , inTheMiddle , atBottom = vertical position relative to the VMS edge
positionXRelativeToPrevious	0..1	toTheRight = alignment to previous
positionYRelativeToPrevious	0..1	alignedOnTheBottomSide = alignment to previous
maxNumberOfRows	1..1	2; number of text lines on a text panel
textCodeListIdentifier	0..1	cz-phrase-code-list-v1.0; Table identifier with codified meaning of phrases displayed on text panel (not all phrases must have codified meaning).

Element PictogramDisplayArea [1..n]

element	card.	content/example
positionXAbsolute	0..1	onLeft , inTheMiddle , onRight = horizontal position relative to the VMS edge
positionYAbsolute	0..1	atTop , inTheMiddle , atBottom = vertical position relative to the VMS edge
positionXRelativeToPrevious	0..1	alignedOnTheBottomSide , toTheRight = alignment to previous
pictogramCodeListIdentifier	0..1	cz-pictogram-code-list-v1.0; Identifier of the table with codified meaning of pictograms displayed on the pictogram (usually it is a table containing the mapping of the meaning of the road sign to DATEX II but additional information may be present.)
pictogramNumberOfColours	0..1	3; for simpleMatrix, red (border), white (symbol), orange (flashes), background black

Element vmsLocation [1]

element	card.	content/example
supplementaryPositionalDescription	1..1	structure; Description of VMS position to lane level including name/number and road direction
pointByCoordinates	1..1	structure; Coordinates in WGS84

pointAlongLinearElement	1..2	structure ; Standard description of the location of a point along a road using the distance from the origin of the element in two variants: Distance from the origin of the named element Distance from the beginning of the GlobalNetwork segment
OpenlrPointAlongLine	0..1	structure ; Optional description with OpenLR

Element supplementaryPositionalDescription [1]

element	card.	content/example
positionOnCarriageway	1..1	Position of VMS on the structure along or above the road: onTheLeft : on the post (of a portal) on the left in the direction of travel inTheCentre : on the gantry above the road (also applies to semi-portals) onTheRight : on the post (of a portal) on the right in the direction of travel
<i>carriageway [1..1] (the following 4 elements are part of the carriageway structure)</i>		
carriageway	1	mainCarriageWay ; always this value (mandatory)
originalNumberOfLanes	0..1	3; Original number of lanes, whenever this information is known
lane\laneNumber	0..1	1; the first lane from the right (0=hard shoulder), determines which lane is controlled by the VMS, if VMS applies to the whole road belt it is omitted
lane\laneUsage	0..1	allLanesCompleteCarriageWay ; if the sign controls the whole road belt, otherwise it is omitted (either laneNumber or laneUsage is used)
<i>roadInformation [1..1] (the following 3 elements are part of the roadInformation structure)</i>		
roadDestination	0..1	Prague; name of the place where the traffic flow is directed
roadName	0..1	Wilsonova; name of the road, street on which VMS "lies"
roadNumber	0..1	D8; number of the road on which the VMS "lies"

Element pointByCoordinates [1]

element	card.	content/example
pointCoordinates\latitude	1	Coordinates of the position of the VMS in WGS84, at the intersection of the centre of the carriageway and the perpendicular line to the position of the post along the road
pointCoordinates\longitude	1	

Element pointAlongLinearElement [1] - mileage

element	card.	content/example
directionRelativeAtPoint	1..1	in the direction or against the direction of stationing (mileage), values aligned opposite
LinearElementByCode\linearElementReferenceModel	1..1	mileage ; description of the position of the mileage
LinearElementByCode\linearElementIdentifier	1	D8; identifiable road name/number
DistanceFromLinearElementStart\distanceAlong	1	2325; distance in metres from the beginning of the road

Element pointAlongLinearElement [0..1] - GlobalNetwork

NDIC - DATEX II VMS Table Publication – Variable Message Signs, Release 1.1rev0

element	card.	content/example
directionRelativeAtPoint	1..1	in the direction or against the direction of the GN network element coding, allowed values aligned opposite
LinearElementByCode\linearElementReferenceModel	1..1	globalNetwork; description of the position relative to the Global Network model (similar to earlier extensions)
LinearElementByCode\linearElementReferenceModelVersion	0..1	21.06; identification of the specific version of the Global Network
LinearElementByCode\linearElementIdentifier	1	145789632; identification of a specific segment from a given Global Network version
PercentageDistanceAlongLinearElement\percentageDistanceAlong	1	28.5; distance in <u>percent</u> from the origin of the segment in the range 0-100, where 100 = 100% (this is the ratio of the length of the section from the start to the given point to the total length of the section)

6 Changelog

version 1.1

- added additional `physicalSupportEnum` values for e.g. bridge mounted VMSs. Added `feedDescription` to link to source uri
- corrected samples `feedType` and headers

version 1.0.0

- Initial format and documentation